Case Study: Distinguishing AI and Real Images Rubric

DS 4002 - Spring 2026 - Author: Jeremy Grossman
Due: TBD

In this assignment, you will use a provided labeled images dataset to train a classifier to predict whether the image is real or AI-generated. You will apply a ConvNeXT model to this data, specifically ConvNeXT_Tiny.

Submission Format:

You have three deliverables to submit for this assignment

1: A 1-2 page PDF summarizing your work, results, and next steps

2: A GitHub repository of your code and outputs

3: A list of references in IEEE citation style

PDF requirements	*Submit as a 1-2 page PDF that contains the following components:* - **Task Choice and Justification:** State whether you chose to train a model for task A (binary classification) or task B (6-class classification) and justify your decision. - **Epoch Number Choice and Justification:** State the number of epochs you chose to train your model for and explain why you chose this number. - **TODO fill-in:** Explain how you filled in the TODO blocks in the code to obtain the results you got. If you used any online sources or AI assistance while writing your code, explain how you used them here. - **Results:** Include the confusion matrix, F1 score, and accuracy (all for the test set) in your PDF. Give a brief analysis of your model's performance. - **Next Steps:** Describe what the next steps could be regarding classifying images as AI or real.

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<tr><td>GitHub repository requirements</td><td>Submit a link to a GitHub repository with the following components:

Firstname_Lastname_AI_Code.ipynb: A Jupyter Notebook based off the given Case_Study_Code.ipynb template with the following components:

Choose either TASK =”A” (binary classification) or TASK = “B” (6-class classification)
Complete the TODOS in the train_one_epoch() function
Complete the TODOS in the evaluate() function
Set the “epochs” variable equal to however many epochs you want to train your model for.
Complete the TODO to create a confusion matrix to visualize your results

Confusion_Matrix.pdf: A PDF file containing the aforementioned confusion matrix you created in your Jupyter Notebook
README.md: Explain what this repository contains very briefly. THis does not need to be overly long
</td></tr>
<tr><td>References</td><td>Submit a PDF with the following:

List of references: Include any online sources, AI sources, or physical sources that you used to help you complete either of the above deliverables.
IEEE citation style
You do NOT need to cite the 2 articles already provided for you
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